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Fraud Detection in Healthcare Insurance Claims using GCN & Explainable AI

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14/08/2026

Introduction

- ▶ Healthcare fraud is a major global problem, The traditional rule based or manual detection struggles with evolving fraud.
- ▶ Medicare scams have resulted in loss of more than 100 million dollars in false claims [2]
- ▶ Healthcare data is highly confidential, as they violate the privacy concerns of users
- ▶ Key challenges: high dimensional claims data and severe class imbalance.
- ▶ WOA and GA can support feature selection and hyperparameter optimization [2].
- ▶ GCN models relationships among providers, beneficiaries, diagnoses, procedures and time rather than treating records independently [4].

Types of Fraud

The following are common healthcare fraud types identified in the literature [5]:

- ▶ **Upcoding:** Billing for a more expensive service than provided.
- ▶ **Phantom Billing:** Billing for services that were never provided.
- ▶ **Unnecessary Care:** Billing for medically unnecessary services.
- ▶ **Double Billing:** Billing multiple times for the same service.
- ▶ **Unbundling:** Separating services that should be billed together.
- ▶ **Wrong Diagnosis:** Using an incorrect diagnosis to justify a claim.
- ▶ **Kickbacks:** Illegal financial benefits for referrals or healthcare decisions.
- ▶ **Self Referral:** Referring patients to a service in which the provider has a financial interest.

Literature Survey : Existing Solutions

S.No.	Paper Details	Techniques Used	Gaps Identified
1	Nabrawi & Alanazi, <i>Fraud detection in healthcare insurance claims using machine learning</i> , <i>Risks</i> , MDPI 2023 [1]	▶ Random Forest ▶ Logistic Regression ▶ Artificial Neural Network (ANN)	▶ No explicit handling of high dimensional data ▶ Limited exploration of optimization based tuning
2	Tubishat et al., <i>Leveraging evolutionary algorithms with a dynamic weighted search space approach for fraud detection in healthcare insurance claims</i> , <i>Knowledge-Based Systems</i> , Elsevier 2025 [2]	▶ Dynamic Weighted Search Space ▶ Whale Optimization Algorithm (WOA) ▶ Support Vector Machine (SVM) ▶ Feature weighting ▶ LLM based synthetic data generation	▶ Nested cross validation is computationally expensive ▶ Synthetic data may not fully represent real world fraud diversity ▶ Extensive evaluation focused mainly on WOA family methods

Literature Survey : Existing Solutions

S.No.	Paper Details	Techniques Used	Gaps Identified
3	Cai et al., <i>WCGAN-GA-RF: Healthcare Fraud Detection via Generative Adversarial Networks and Evolutionary Feature Selection</i> , Information, MDPI 2026 [3]	▶ WCGAN ▶ Synthetic minority oversampling ▶ Genetic Algorithm (GA) ▶ Feature selection ▶ Random Forest	▶ Precision recall trade off ▶ Lower recall for minority fraud class ▶ Validated on a single regional dataset
4	Hong et al., <i>Health insurance fraud detection based on multi channel heterogeneous graph structure learning</i> , Heliyon Elsevier, 2024 [4]	▶ Heterogeneous graph construction ▶ Topology graph & Feature graph & Semantic graph ▶ Multi channel GCN ▶ Attention based fusion	▶ Medical A and Medical B datasets are confidential ▶ Limited availability of publicly reproducible data ▶ Explainable AI is not the central focus

Motivation & Problem Statement

Motivation

- ▶ Healthcare insurance fraud causes substantial financial losses and affects the efficient utilization of healthcare resources.
- ▶ Healthcare claims contain complex relationships among **patients, departments, medicines, and time** that are difficult to capture using conventional machine learning models [4].
- ▶ Fraudulent behavior may be revealed not only by individual claim features, but also through **structural, feature based, and semantic relationships** between nodes [4].
- ▶ Fraud detection models should also provide an understandable explanation of **why a provider is classified as potentially fraudulent**.

Motivation & Problem Statement

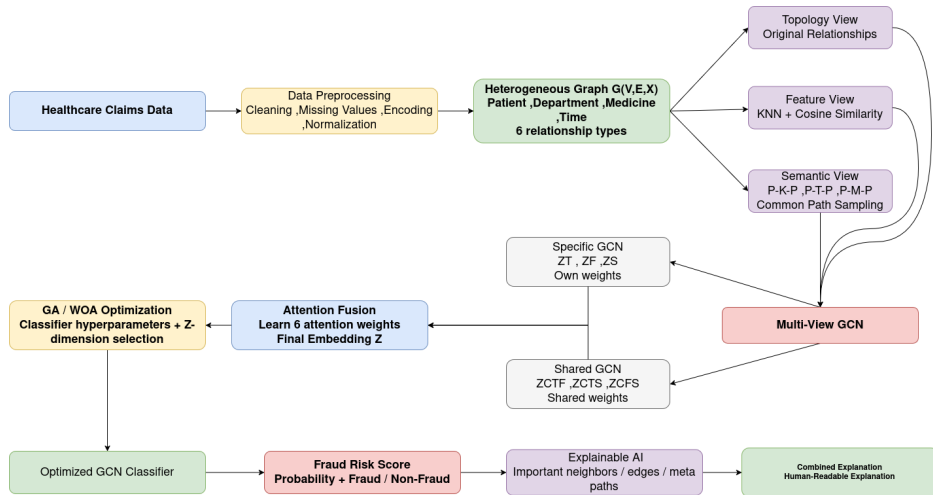
Problem Statement

- ▶ Healthcare fraud detection is challenging because of **complex heterogeneous relationships, high dimensional features, and imbalanced fraud & non fraud classes**.
- ▶ Conventional classifiers may identify fraudulent providers but often fail to capture the underlying graph relationships and provide limited reason level explanations.
- ▶ Hence, there is a need for an integrated framework that combines:
 - ▶ Multi view heterogeneous GCN
 - ▶ Attention based representation learning
 - ▶ Evolutionary optimization
 - ▶ Explainable AI

Objectives

- ▶ Develop a **heterogeneous multi view GCN** for provider level healthcare insurance fraud detection.
- ▶ Integrate **GA/WOA optimization** for feature/embedding selection and model parameter tuning.
- ▶ Improve fraud classification using an **optimized GCN** and evaluate using Accuracy, Precision, Recall, F1-score and AUC.
- ▶ Use **Explainable AI** to identify important features, nodes and edges and generate **human readable fraud reasons**.

Proposed Methodology



Dataset 1: Healthcare Provider Fraud Detection Analysis

- ▶ **Source:** Kaggle Dataset
- ▶ **Domain:** Medicare healthcare provider fraud
- ▶ Contains **Inpatient Claims, outpatient claims, and Beneficiary Details.**
- ▶ Target: **Potentially fraudulent provider.**
- ▶ Useful features include provider information, claim amounts, diagnosis information, beneficiary utilization, and service patterns.
- ▶ Suitable for **provider level fraud detection** and graph construction.

Dataset 2: NHIS Healthcare Claims and Fraud Dataset

- ▶ **Source:** Kaggle Dataset
- ▶ Contains patient demographics, admission details, diagnosis, treatment, and billing information.
- ▶ Provides explicit **fraud & non fraud annotations**.
- ▶ Fraud scenarios include **Fake Treatment** and **Phantom Billing**.
- ▶ Suitable for **claim level fraud detection**.

Expected Timeline

Review 1	Review 2	Review 3	Final Review
▶ Problem Identification ▶ Literature Survey ▶ Identify Research Gaps ▶ Formulate Problem Statement ▶ Identify Suitable Dataset	▶ Dataset Collection ▶ Data Cleaning and Preprocessing ▶ Heterogeneous Graph Construction ▶ Construct Topology View & Feature View & Semantic View & Multi-View GCN ▶ Initial Model Training ▶ GA/WOA Optimization	▶ Hyperparameter Optimization ▶ Feature/Embedding Optimization ▶ Attention Fusion ▶ Fraud Classification ▶ Explainable AI integration ▶ Generate Explanations ▶ Performance Evaluation	▶ Final Optimized Model ▶ Complete Experimental Results ▶ Baseline Comparison ▶ Ablation Analysis ▶ XAI Results ▶ Final Conclusions

References

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Thank You